

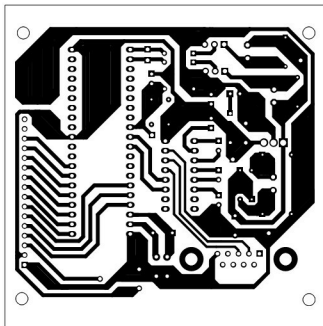


Fig. 2: Actual-size, single-side PCB for the SMS-based automation system

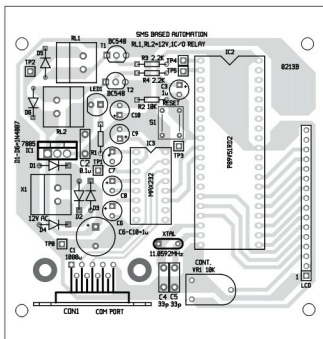


Fig. 3: Component layout for the PCB

Similarly, you can switch on the bulb by sending SMS 'lgton' and turn it off by sending SMS 'lgtof'. GSM modem used in this project is SIM300 V0.7. The message received by the GSM module is also indicated on the LCD display.

Construction and testing

An actual-size, single-side PCB for

the SMS-based automation system is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC base for the microcontroller. Before inserting the IC, check the supply voltage. Using preset VR1, set the LCD contrast for proper display.

To test the circuit for proper functioning, insert SIM in the GSM module and power it 'on.' Red and green LEDs of the GSM module indicate power-'on' and signal reception. The GSM module is connected to the circuit. Make a call on the SIM in the module and check whether the module rings.

To further test the circuit, follow the test points mentioned in the table.

Software

Source program for microcontrollers is written in 'C' language and compiled using Keil μ Vision4 compiler. The generated hex code is burnt into the microcontroller by using a suitable programmer. •

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